

Equation of motion

$$\bar{h}[k+1] = \bar{h}[k] + a_h[k]\{f_{dh}[k] + f_T[k]\} + \delta\bar{h}_D[k],$$

Motion error

$$\delta\bar{h}[k+1] = \delta\bar{h}[k] + \Delta\bar{h}_d[k] - a_h[k]\{f_{dh}[k] + f_T[k]\} - \delta\bar{h}_D[k],$$

$$\delta\bar{h}[k] = \bar{h}_d[k] - \bar{h}[k], \quad \Delta\bar{h}_d[k] = \bar{h}_d[k+1] - \bar{h}_d[k]$$

Measurement

(d -step delay)

$$\delta\bar{h}_m[k] = \delta\bar{h}[k-d] + n_h[k], \quad \textbf{Control objective : } \delta\bar{h}[k+1] = \lambda_c \delta\bar{h}[k], \quad 0 \leq \lambda_c < 1$$

Ideal control effort (no measurement delay), and the resulting motion error:

$$\bar{f}_{dh}[k] = a_h^{-1}[k]\{\Delta\bar{h}_d[k] + (1 - \lambda_c)\delta\bar{h}[k] - \delta\bar{h}_D[k]\}$$

$$\delta\bar{h}[k+1] = \delta\bar{h}[k] + \Delta\bar{h}_d[k] - a_h[k]\{\bar{f}_{dh}[k] + f_T[k]\} - \delta\bar{h}_D[k] = \lambda_c \cdot \delta\bar{h}[k] - a_h[k]f_T[k]$$

d-step measurement delay:

$$\delta\bar{h}[k] = \delta\bar{h}[k-d] + \sum_{i=1}^d \Delta\bar{h}_d[k-i] - \sum_{i=1}^d a_h[k-i]\{f_{dh}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta\bar{h}_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\begin{aligned} \bar{f}_{dh}[k] = a_h^{-1}[k] \Big\{ & \Delta\bar{h}_d[k] - \delta\bar{h}_D[k] + (1 - \lambda_c) \sum_{i=1}^d \left[\Delta\bar{h}_d[k-i] - \delta\bar{h}_D[k-i] \right] \\ & + (1 - \lambda_c) \left[\delta\bar{h}[k-d] - \sum_{i=1}^d a_h[k-i]f_{dh}[k-i] \right] \Big\} \end{aligned}$$

$$\delta\bar{h}[k+1] = \lambda_c \cdot \delta\bar{h}[k] - \left\{ a_h[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i]f_T[k-i] \right\}$$

Implementable control effort

(disturbance not estimated)

$$\boxed{f_{dh}[k] = \hat{a}_h^{-1}[k] \left\{ \Delta\bar{h}_d[k] + (1 - \lambda_c) \left[\sum_{i=1}^d \Delta\bar{h}_d[k-i] + \delta\bar{h}_m[k] - \sum_{i=1}^d \hat{a}_h[k-i]f_{dh}[k-i] \right] \right\}}$$

Motion error

$$\delta\bar{h}[k+1] = \lambda_c \cdot \delta\bar{h}[k] - a_h[k] [f_{dh}[k] - \bar{f}_{dh}[k]] - \left\{ a_h[k] f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i] f_T[k-i] \right\}$$

$$\begin{aligned} a_h[k] f_{dh}[k] - a_h[k] \bar{f}_{dh}[k] &= e_{a_h}[k] f_{dh}[k] + (1 - \lambda_c) n_h[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_h[k-i] - \hat{a}_h[k-i]) f_{dh}[k-i] \\ &\quad + \delta\bar{h}_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta\bar{h}_D[k-i] \end{aligned}$$

Gain-error back-step:

$$\begin{aligned} a_h[k-i] - \hat{a}_h[k-i] &\approx e_{a_h}[k] - (\bar{h}_d[k] - \bar{h}_d[k-i]) (a'_h[k] - \hat{a}'_h[k]) \approx e_{a_h}[k] \\ a_h[k] f_{dh}[k] - a_h[k] \bar{f}_{dh}[k] &\approx e_{a_h}[k] \underbrace{\left\{ f_{dh}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dh}[k-i] \right\}}_{F_{dh}[k]} \\ &\quad + \delta\bar{h}_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta\bar{h}_D[k-i] + (1 - \lambda_c) n_h[k] \end{aligned}$$

$\begin{aligned} \delta\bar{h}[k+1] &= \lambda_c \cdot \delta\bar{h}[k] - F_{dh}[k] \cdot e_{a_h}[k] - \delta\bar{h}_D[k] - (1 - \lambda_c) \sum_{i=1}^d \delta\bar{h}_D[k-i] - \varepsilon_h[k] \\ \varepsilon_h[k] &:= a_h[k] f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i] f_T[k-i] + (1 - \lambda_c) n_h[k] \end{aligned}$

Gain model

($a_o = \Delta t / (\gamma_N R)$ = far-field gain, $\bar{h}_d = h_d / R$; a_h has units 1/pN)

$$a_h(\bar{h}) = a_o [1 - e^{-b\varphi(\bar{h})}], \quad \varphi(1) = 0, \quad \varphi(\bar{h} \rightarrow \infty) = \infty, \quad \varphi' > 0$$

$$a'_h(\bar{h}) := \frac{da_h}{d\bar{h}} = \varphi'(\bar{h}) b a_o e^{-b\varphi(\bar{h})} = \varphi'(\bar{h}) b [a_o - a_h(\bar{h})]$$

$$B[k] := b \varphi'(\bar{h}_d[k]) \quad \implies \quad a'_h[k] = B[k] [a_o - a_h[k]] \quad (b \text{ is a state, } B[k] \text{ is not})$$

Shape function

$$\varphi(\bar{h}) = \ln \bar{h} \quad \implies \quad \varphi'(\bar{h}) = \frac{1}{\bar{h}}, \quad B[k] = \frac{b}{\bar{h}_d[k]}, \quad a_h(\bar{h}) = a_o [1 - \bar{h}^{-b}]$$

$\begin{aligned} a_h(\bar{h}) &= a_o [1 - e^{-b\varphi(\bar{h})}], \\ \varphi(\bar{h}) &= \ln \bar{h}, \end{aligned}$	$\begin{aligned} a'_h[k] &= B[k] [a_o - a_h[k]], \\ B[k] &= b \varphi'(\bar{h}_d[k]) = b / \bar{h}_d[k]. \end{aligned}$	$b[k+1] = b[k],$
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Local exponent demanded by the true curve

The gain law asserts that the normalised gain deficit is a pure exponential in φ :

$$\Psi(\bar{h}) := 1 - \frac{a_h(\bar{h})}{a_o} = e^{-b\varphi(\bar{h})}, \quad \text{true value} \quad \Psi_{\text{true}}(\bar{h}) = 1 - \frac{1}{c(\bar{h})} = \frac{c-1}{c}$$

so b is *by construction* a slope in $(\ln \Psi, \varphi)$ coordinates. Reading that slope off the true curve gives the exponent the truth demands at each height:

$$b = -\frac{d \ln \Psi}{d\varphi} \implies b_{\text{eff}}(\bar{h}) := -\frac{1}{\varphi'(\bar{h})} \frac{d}{d\bar{h}} \ln \frac{c-1}{c} = -\frac{\bar{h} c'(\bar{h})}{c(\bar{h}) [c(\bar{h}) - 1]}$$

Were the shape exact, b_{eff} would be constant. **Its variation is the model error**, and it is the only quantity that can set the prior on b : b is a parameter of an approximate shape, not a physical constant, so “how wrong is $\hat{b}[0] = 1$ ” has no meaning beyond “how far does b_{eff} stray from 1 over the reachable domain”.

Two limits pin $b = 1$

(both read off the published asymptotics; c itself is never needed at run time)

$$\begin{aligned} \bar{h} \rightarrow 1: \quad c \rightarrow \frac{K}{\bar{h}-1} &\Rightarrow \ln \Psi \simeq -\frac{\bar{h}-1}{K}, \quad \varphi \simeq \bar{h} - 1 &\Rightarrow b_{\text{eff}} \rightarrow 1/K \\ \bar{h} \rightarrow \infty: \quad c-1 = A\bar{h}^{-1} + A^2\bar{h}^{-2} + \dots &\Rightarrow \Psi = A\bar{h}^{-1} + O(\bar{h}^{-3}), \quad \ln \Psi = \ln A - \varphi &\Rightarrow b_{\text{eff}} \rightarrow 1 \end{aligned}$$

The far-field anchor is coefficient-free (A is additive in $\ln \Psi$, so it drops out of the slope), and the $O(\bar{h}^{-2})$ terms cancel identically in $\Psi = (c-1)/c$ — which is why the exponential shape is *more* accurate far from the wall than the power law ($b_{\text{eff}} = 0.9983$ vs $p_{\text{eff}} = 1.0042$ at $\bar{h} = 22.2$). The near-wall anchor is *weaker* than the power law’s: it returns $1/K$, so it pins $b = 1$ only because Brenner lubrication carries unit coefficient $K = 1$, whereas the power law’s near-wall anchor gives 1 for any K .

Acceptance criterion

(evaluated offline once on the exact correction polynomials; the controller carries only the resulting number)

shape	exponent	$\perp \quad (c_{\perp})$		$\parallel \quad (c_{\parallel})$	
		all $\bar{h}$	$\bar{h} \geq 2$	all $\bar{h}$	$\bar{h} \geq 2$
power law $c-1 = K(\bar{h}-1)^{-p}$	p	0.034	0.024	0.994	0.283
exponential $\Psi = \bar{h}^{-b}$	b	0.094	0.082	0.778	0.034

Tabulated: $\sup |\text{exponent} - 1|$. The exponential shape trades $\perp$ accuracy ($2.8\text{--}3.4\times$ worse) for $\parallel$ viability ($8.3\times$ better over the working range), which is what lets one gain law serve all three axes; the power law has no near-wall anchor for $\parallel$ at all, since $c_{\parallel}$ stays finite at contact.

Since b is modelled as constant, $P_{55}[0]$ is the entire budget the filter may spend on it ($Q_{55} = 0$ and $\mathbf{F}_e$ row 5 = $[00001]$ give $\mathbb{E}[(\hat{b}_{\infty} - \hat{b}_0)^2] = P_{55}[0] - P_{55}[\infty]$). The shape is therefore acceptable exactly when that budget covers its own mismatch:

$$\boxed{\sup_{\bar{h}} |b_{\text{eff}}(\bar{h}) - 1| \lesssim \sqrt{P_{55}[0]} \iff Q_{55} = 0 \text{ is honest}}$$

Current status. $\sqrt{P_{55}[0]} = 0.10$ against 0.0942 for $\perp$: the criterion is met with a factor 1.06, i.e. no margin. (The power law on $\perp$ sat at 0.035 against 0.034 — the same tightness.) A shape whose exponent varied by, say, 0.01 would let the prior shrink by an order of magnitude and turn b from a nuisance parameter into an identified one.

Why a non-zero Q_{55} is the wrong repair

If b_{eff} varies with height, the true exponent moves as

$$b_{\text{true}}[k+1] - b_{\text{true}}[k] = \frac{db_{\text{eff}}}{d\bar{h}} \Delta\bar{h}_d[k]$$

which is *deterministic and commanded*: $\Delta\bar{h}_d$ is the trajectory increment, known exactly and noise-free. Such a term belongs in the prediction as a feedforward on row 5, not in Q . Modelling it as a random walk, $Q_{55}[k] = \kappa_b^2 \Delta\bar{h}_d^2[k]$ with $\kappa_b := \sup |db_{\text{eff}}/d\bar{h}| = 0.032$ ($\perp$, $\bar{h} \geq 2$), does keep the correct trajectory gating — $Q_{55} = 0$ while holding, since b_{true} does not move when the probe does not — but it diffuses as $\sqrt{N}$ where the true drift accumulates as N : over the canonical descent $\bar{h} : 22.2 \rightarrow 2$ in $N = 1600$ steps it supplies $\kappa_b \sqrt{N} \Delta\bar{h}_d = 0.016$, while b_{eff} actually travels $0.9983 \rightarrow 0.9184 = 0.080$ — short by $5\times$ even with κ_b taken as a supremum. Inflating κ_b only contaminates the hold phases without restoring the shape. The exact feedforward needs $db_{\text{eff}}/d\bar{h}$, i.e. c itself, which this gain law exists to avoid; estimating it instead adds a state and reopens the Taylor-ladder regress the two-parameter law was introduced to close. The criterion above is therefore stated on the shape, never on Q .

State

($\mathbf{y}$ carries $\delta\bar{h}[k-d]$; a_h and b are unknown)

$$\delta\bar{h}_j[k] := \delta\bar{h}[k-d+j-1], \quad j = 1, \dots, d+1; \quad \mathbf{x}[k] = [\delta\bar{h}_1, \delta\bar{h}_2, \delta\bar{h}_3, a_h, b]^\top[k] \quad (d=2)$$

Tracking error

(boxed motion error, $\delta\bar{h}_D \equiv 0$ hereafter)

$$\delta\bar{h}_3[k+1] = \lambda_c \delta\bar{h}_3[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k]$$

Gain recursion

(the gain is evaluated at the true point, which trails the command by $\delta\bar{h}_3$)

$$\bar{h}[k] = \bar{h}_d[k] - \delta\bar{h}_3[k]$$

$$a_h[k] = a_h(\bar{h}_d[k]) - a'_h[k] \delta\bar{h}_3[k]$$

$$a_h[k+1] = a_h(\bar{h}_d[k+1]) - a'_h[k] \delta\bar{h}_3[k+1] = a_h(\bar{h}_d[k]) + a'_h[k] \Delta\bar{h}_d[k] - a'_h[k] \delta\bar{h}_3[k+1]$$

Subtracting,

$$a_h[k+1] = a_h[k] + a'_h[k] \left\{ \underbrace{\Delta\bar{h}_d[k] - (\delta\bar{h}_3[k+1] - \delta\bar{h}_3[k])}_{\text{step taken by the true position}} \right\}$$

and substituting $\delta\bar{h}_3[k+1]$,

$$a_h[k+1] = a_h[k] + a'_h[k] \left\{ \Delta\bar{h}_d[k] + (1 - \lambda_c) \delta\bar{h}_3[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k] \right\}$$

True state equation

$$\begin{aligned}
\delta\bar{h}_1[k+1] &= \delta\bar{h}_2[k] \\
\delta\bar{h}_2[k+1] &= \delta\bar{h}_3[k] \\
\delta\bar{h}_3[k+1] &= \lambda_c \delta\bar{h}_3[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k] \\
a_h[k+1] &= a_h[k] + a'_h[k] \{ \Delta\bar{h}_d[k] + (1 - \lambda_c) \delta\bar{h}_3[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k] \} \\
b[k+1] &= b[k], \quad a'_h[k] = B[k] [a_o - a_h[k]]
\end{aligned}$$

Measurement

(every length, σ_{n_h} included, is already normalized by R)

$$\nabla_d \bar{h}_d[k] := \bar{h}_d[k] - \bar{h}_d[k-d]$$

$$y_1[k] = \delta\bar{h}_m[k] = \delta\bar{h}_1[k] + \underbrace{n_h[k]}_{r_1[k]}$$

$$y_2[k] = a_{hm}[k] = a_h[k-d] + \underbrace{n_a[k]}_{r_2[k]}$$

Delay back-off

$$a_h[k-d] = a_h[k] - a'_h[k] \nabla_d \bar{h}_d[k] \quad \implies \quad y_2[k] = a_h[k] - a'_h[k] \nabla_d \bar{h}_d[k] + n_a[k]$$

Linearization

(Jacobian at the estimate; partials at fixed a_h , since a_h and b are independent states)

$$\frac{\partial y_2}{\partial a_h} = 1 + \nabla_d \bar{h}_d[k] \hat{B}[k] \longrightarrow 1, \quad \frac{\partial y_2}{\partial b} = -\nabla_d \bar{h}_d[k] \frac{\hat{a}'_h[k]}{\hat{b}}$$

Output equation

($\mathbf{H}$ is the Jacobian; the second row is nonlinear in $\mathbf{x}$, so $\mathbf{H}\mathbf{x}$ is its linearization, not an identity)

$$\mathbf{y}[k] = \mathbf{H}[k] \mathbf{x}[k] + \mathbf{r}[k], \quad \mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\nabla_d \bar{h}_d[k] \hat{a}'_h[k]/\hat{b} \end{bmatrix}, \quad \mathbf{r}[k] = \begin{bmatrix} n_h[k] \\ n_a[k] \end{bmatrix}$$

Estimation error

$$e_{\delta\bar{h}_j} = \delta\bar{h}_j - \hat{\delta\bar{h}}_j \quad (j = 1, 2, 3), \quad e_{a_h} = a_h - \hat{a}_h, \quad e_b = b - \hat{b}, \quad \mathbf{e} = [e_{\delta\bar{h}_1}, e_{\delta\bar{h}_2}, e_{\delta\bar{h}_3}, e_{a_h}, e_b]^\top$$

Innovation

($\hat{a}'_h = \hat{B}[k][a_o - \hat{a}_h]$, $\hat{B}[k] = \hat{b} \varphi'(\bar{h}_d[k])$; the second line mixes evaluation points, which is what makes it exact)

$$e_{y_1} = y_1 - \delta\hat{\bar{h}}_1 = e_{\delta\bar{h}_1} + n_h$$

$$\begin{aligned}
a'_h - \hat{a}'_h &= \varphi'(\bar{h}_d) \{b[a_o - a_h] - \hat{b}[a_o - \hat{a}_h]\} = -\hat{B}[k] e_{a_h} + \frac{a'_h[k]}{\hat{b}} e_b \\
e_{y_2} &= y_2 - (\hat{a}_h - \hat{a}'_h \nabla_d \bar{h}_d) = (1 + \nabla_d \bar{h}_d \hat{B}[k]) e_{a_h} - \nabla_d \bar{h}_d \frac{\hat{a}'_h[k]}{\hat{b}} e_b + n_a \\
&\longrightarrow e_{a_h} - \nabla_d \bar{h}_d \frac{\hat{a}'_h[k]}{\hat{b}} e_b + n_a
\end{aligned}$$

Estimator

($\hat{a}_h$ swept by the step the estimator knows)

$$\begin{aligned}
\delta \hat{h}_1[k+1] &= \delta \hat{h}_2 + \ell_{11} e_{y_1} + \ell_{12} e_{y_2} \\
\delta \hat{h}_2[k+1] &= \delta \hat{h}_3 + \ell_{21} e_{y_1} + \ell_{22} e_{y_2} \\
\delta \hat{h}_3[k+1] &= \lambda_c \delta \hat{h}_3 + \ell_{31} e_{y_1} + \ell_{32} e_{y_2} \\
\hat{a}_h[k+1] &= \hat{a}_h + \hat{a}'_h \{ \Delta \bar{h}_d + (1 - \lambda_c) \delta \hat{h}_3 \} + \ell_{41} e_{y_1} + \ell_{42} e_{y_2} \\
\hat{b}[k+1] &= \hat{b} + \ell_{51} e_{y_1} + \ell_{52} e_{y_2}
\end{aligned}$$

Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\delta \bar{h}_1}[k+1] &= e_{\delta \bar{h}_2} - \ell_{11} e_{y_1} - \ell_{12} e_{y_2} \\
e_{\delta \bar{h}_2}[k+1] &= e_{\delta \bar{h}_3} - \ell_{21} e_{y_1} - \ell_{22} e_{y_2} \\
e_{\delta \bar{h}_3}[k+1] &= \lambda_c e_{\delta \bar{h}_3} - \underbrace{F_{dh}}_{q_3} e_{a_h} - \ell_{31} e_{y_1} - \ell_{32} e_{y_2} \\
e_{a_h}[k+1] &= \left(1 - [\Delta \bar{h}_d + (1 - \lambda_c) \delta \hat{h}_3] \hat{B} + \hat{a}'_h F_{dh} \right) e_{a_h} + (1 - \lambda_c) \hat{a}'_h e_{\delta \bar{h}_3} + [\Delta \bar{h}_d + (1 - \lambda_c) \delta \hat{h}_3] \frac{\hat{a}'_h}{\hat{b}} e_b \\
&\quad + \underbrace{a'_h \varepsilon_h}_{q_4} - \ell_{41} e_{y_1} - \ell_{42} e_{y_2} \\
e_b[k+1] &= e_b - \ell_{51} e_{y_1} - \ell_{52} e_{y_2}
\end{aligned}$$

Error state equation

($\mathbf{q} = [0, 0, q_3, q_4, 0]^\top$, $q_3 = -\varepsilon_h$, $q_4 = a'_h[k] \varepsilon_h = -a'_h[k] q_3$; entries evaluated at the current estimate)

$$\boxed{
\begin{aligned}
\mathbf{e}[k+1] &= \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H}[k] \mathbf{e}[k] + \mathbf{q}[k] \\
\mathbf{F}_e[k] &= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh} & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{a}'_h & 1 - [\Delta \bar{h}_d + (1 - \lambda_c) \delta \hat{h}_3] \hat{B} + \hat{a}'_h F_{dh} & [\Delta \bar{h}_d + (1 - \lambda_c) \delta \hat{h}_3] \hat{a}'_h / \hat{b} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\end{aligned}
}$$

Process noise Q

(only the current white thermal step enters Q : the past thermal steps in ε_h are cross-step correlated, and n_h already sits in R_1)

$$q_3 = -a_h[k] f_T[k], \quad q_4 = a'_h[k] a_h[k] f_T[k] = -a'_h[k] q_3$$

$$\gamma(\bar{h}) = \gamma_N c(\bar{h}), \quad \sigma_{f_T}^2 = \frac{4k_B T \gamma(\bar{h})}{\Delta t}, \quad a_h = \frac{\Delta t}{\gamma(\bar{h}) R}$$

$$\text{Var}(a_h f_T) = a_h^2 \sigma_{f_T}^2 = \frac{4k_B T}{R} a_h[k]$$

$Q_{33}[k] = \frac{4k_B T}{R} a_h[k],$ $Q_{44}[k] = a_h'^2[k] Q_{33}[k],$	$Q_{34}[k] = Q_{43}[k] = -a_h'[k] Q_{33}[k],$ $Q_{55} = 0 \quad (b \text{ constant});$
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All other entries vanish; the gain block is rank one, since $q_4 = -a_h'[k] q_3$.

Measurement noise $R = \text{diag}(R_1, R_2)$

(a_{cov} = EWMA coefficient, IF_{var} = $\text{MA}(d)$ inflation factor of $\hat{\sigma}_{\delta\bar{h}_r}^2$)

$$R_1 = \sigma_{n_h}^2$$

$$\sigma_{\delta\bar{h}_r}^2[k] = C_{dpmr} \frac{4k_B T}{R} a_h[k] + C_n \sigma_{n_h}^2 = C_{dpmr} \frac{4k_B T}{R} (a_h[k] + \xi), \quad \xi := \frac{C_n}{C_{dpmr}} \frac{\sigma_{n_h}^2}{4k_B T/R}$$

$$a_{hm}[k] = \frac{\hat{\sigma}_{\delta\bar{h}_r}^2[k] - C_n \sigma_{n_h}^2}{C_{dpmr} (4k_B T/R)}$$

$$\text{Var}(\hat{\sigma}_{\delta\bar{h}_r}^2) = K_{var} \text{IF}_{var} (\sigma_{\delta\bar{h}_r}^2)^2, \quad K_{var} = \frac{2a_{cov}}{2 - a_{cov}}$$

$$\boxed{R_2[k] = K_{var} \text{IF}_{var} (a_h[k] + \xi)^2 + d Q_{44}[k]}$$

